

Generative-AI Exposure in Sub-Saharan African Labour Markets

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Abstract

I measure occupational exposure to generative AI across the workforce of four Sub-Saharan African economies, using harmonised LSMS-ISA microdata for 23,444 workers in Nigeria, Tanzania, Uganda, and Ethiopia. Exposure is low and highly concentrated. The pooled mean is 0.088 on a zero-to-one scale, a third of workers face near-zero exposure, and the most exposed tenth of workers carry about a third of all exposure. The dividing line is sectoral, not contractual. Agricultural workers, who make up 59 percent of the pooled workforce, average 0.05, while non-agricultural wage and self-employed workers are equally exposed at 0.14, and formal and informal wage workers do not differ. The same structure scales up. Across 32 African economies, country-level exposure falls steeply with the informal employment rate and the agricultural self-employment share, and rises with income per capita. Exposure in these economies is not merely lower on average but structurally confined to a small, urban, white-collar minority.

Keywords: generative AI, occupational exposure, informality, structural transformation, Sub-Saharan Africa, LSMS-ISA.

JEL codes: J21, J24, O33, O55.

1 Introduction

The rapid diffusion of generative AI has prompted a wave of estimates of how exposed jobs are to the new technology (Felten et al., 2021; Eloundou et al., 2024; Gmyrek et al., 2023). Most of this evidence concerns high-income labour markets, and the cross-country work that

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does cover developing economies builds occupational exposure from adult-skills surveys such as PIAAC and STEP (Pizzinelli et al., 2023; Cazzaniga et al., 2024). Those instruments, by design, sample the formal and largely urban workforce. They do not represent the subsistence farmers, own-account traders, and casual labourers who make up the majority of employment in low-income Sub-Saharan Africa. As a result, the question of who in these economies is actually exposed, and how exposure is distributed across occupational structure, has not been answered with representative data.

This paper takes a different data source as its starting point. The Living Standards Measurement Study Integrated Surveys on Agriculture (LSMS-ISA) are nationally representative household surveys that record the main-job occupation of every working member, including the agricultural and informal majority (World Bank, 2024). I assemble the LSMS-ISA wave with coded occupations for four countries, Nigeria (2023), Tanzania (2020), Uganda (2019), and Ethiopia (2021), and map each worker’s occupation to a task-level measure of generative-AI exposure through a transparent occupation crosswalk. The pooled sample contains 23,444 workers.

This paper is descriptive and its contribution is in three parts. First, it measures the level and distribution of AI exposure across a representative Sub-Saharan workforce and finds exposure low and highly concentrated. The pooled mean is 0.088 on a zero-to-one automation scale, the median worker’s exposure is 0.076, and the most exposed ten percent of workers carry about a third of total exposure. Second, it shows that the concentration is occupational and sectoral rather than contractual. The gap is between agriculture and everything else, not between formal and informal jobs such that agricultural workers average 0.05, while non-agricultural wage employees and own-account workers are both at 0.14, and formality among wage workers makes no difference. Third, it shows that the within-country pattern scales up. In a cross-section of 32 African economies, aggregate exposure falls with the informal employment rate and with agricultural self-employment, and rises with income, such that the four countries observed at the worker level sit on a regional gradient.

These results qualify the view that developing countries are less exposed to AI (Pizzinelli et al., 2023; Cazzaniga et al., 2024). Using data that include the informal and agricultural workforce, I find that exposure in these economies is not simply lower on average but is structurally confined to a small, urban, white-collar minority, with the large agricultural and own-account majority insulated by the nature of their work. This is a measurement and mapping exercise. I do not estimate the effect of AI on employment.

1.1 Related literature

The paper connects to three strands of work. The first measures the exposure of jobs to automation and to AI. The task framework of Autor et al. (2003) and Acemoglu and Restrepo (2019, 2022) treats an occupation as a bundle of tasks and asks which of those tasks a technology can perform. A series of occupation-level indices apply this idea, from the early computerisation probabilities of Frey and Osborne (2017) to the AI occupational exposure scores of Felten et al. (2021) and Webb (2020) and the large-language-model exposure of Eloundou et al. (2024). The exposure measure used here follows this tradition, aggregating task-level capability ratings for generative models to occupations (Althoff and Reichardt, 2026). A parallel literature estimates the realised effects of generative AI on productivity and on labour demand, mostly in high-income or online settings (Brynjolfsson et al., 2023; Noy and Zhang, 2023; Hui et al., 2023). This paper does not estimate effects but instead maps exposure.

The second strand carries exposure measurement across countries and raises the concern that AI may widen gaps between nations (Alonso et al., 2020). Cross-country estimates consistently find that lower-income economies are less exposed, because they employ fewer workers in the professional and clerical occupations that AI reaches (Pizzinelli et al., 2023; Cazzaniga et al., 2024; Gmyrek et al., 2023). These estimates rest on adult-skills surveys such as PIAAC and STEP that sample the formal workforce in a limited set of countries, so they describe the exposed minority rather than the whole labour force. This paper uses household survey data that include the agricultural and informal majority, and asks how exposure is distributed once that majority is in the sample.

The third strand concerns informality and structural transformation. Informal and own-account employment dominates the labour market in low-income economies (La Porta and Shleifer, 2014), and a large share of work remains in low-productivity agriculture (Gollin et al., 2014). These features are usually studied as obstacles to development. Here they appear instead as the reason AI exposure is so concentrated, since the agricultural and own-account work that absorbs most of the workforce is made up of tasks current models cannot perform.

2 Data and measurement

2.1 Surveys and occupation coding

For each of four countries I use the LSMS-ISA wave that records coded occupations, namely the Nigeria General Household Survey-Panel Wave 5 (2023), the Tanzania National Panel

Survey Round 5 (2020), the Uganda National Panel Survey (2019), and the Ethiopia Socioeconomic Panel Survey Wave 5 (2021). Each survey records the main-job occupation of working household members. Nigeria, Tanzania, and Uganda code occupation at the four-digit ISCO-08 level (International Labour Organization, 2012). Ethiopia records occupation at the one-digit major-group level. Occupation coverage is complete among workers in Nigeria and Uganda. In Tanzania the labour module codes occupation only for non-farm work, so I add subsistence farmers, identified from the agricultural-labour question, at the agricultural occupation level. The Ethiopian sample is smaller and selected, so I treat its level with caution and rely on it mainly for the major-group profile.

2.2 Exposure measure and crosswalk

I measure occupational exposure to generative AI with task-level capability ratings that score each O*NET task for whether current generative models can perform it (Althoff and Reichardt, 2026). Ratings are aggregated to occupations using task weights, yielding an automation-exposure score on a zero-to-one scale for each O*NET-SOC occupation. To attach this score to survey workers, I build a crosswalk from each survey’s occupation classification to O*NET-SOC, mapping ISCO-88 and ISCO-08 to the 2010 Standard Occupational Classification and then to O*NET-SOC with the standard correspondence tables. Where one source occupation maps to several O*NET occupations, I take the unweighted mean. As a check, I recompute exposure from an independent occupation-level automation index. It yields the same cross-occupational ranking, and my conclusions concern that ranking rather than the cardinal scale.

2.3 Harmonised variables

For each worker, I record sex, age, urban or rural residence, the four-digit occupation and its ISCO major group, and an employment-status category. Status is anchored on the occupation. Agricultural workers are identified from ISCO major group 6 together with agricultural elementary occupations, and the remaining workers are split into wage employees and own-account or self-employed workers using each survey’s main-job employer question. For wage workers, I additionally construct an informality flag from the presence of a written contract, pension contributions, or income-tax withholding, depending on what each survey records. The resulting pooled file covers 23,444 workers (7,708 in Nigeria, 7,481 in Tanzania, 7,671 in Uganda, and 584 in Ethiopia). Table 1 describes the sample. The four workforces differ in the ways the structural account predicts. The agricultural share ranges from 28 percent in the more urban Nigerian sample to 78 percent in Tanzania, and mean exposure moves

inversely with it.

Table 1: Sample composition by country

Country	<i>N</i>	% female	Mean age	% urban	% agriculture	Mean exposure
Nigeria 2023	7,708	50.5	26.0	19.0	28.4	0.122
Tanzania 2020	7,481	50.5	39.3	3.5	77.9	0.083
Uganda 2019	7,671	65.2	26.6	22.4	72.8	0.055
Ethiopia 2021	584	38.0	26.5	44.5	22.4	0.144
All (pooled)	23,444	55.0	30.5	15.8	58.6	0.088

Workers with a coded main-job occupation. Mean exposure is the task-weighted automation-exposure score on a zero-to-one scale. The agriculture column is the share of workers in agricultural occupations.

3 The structure of AI exposure

Generative-AI exposure reaches only a small part of the workforce in the sample. The pooled mean automation-exposure score is 0.088 and the median is 0.076. One third of workers (33.7 percent) have near-zero exposure below 0.05, and two thirds (66.1 percent) are in agricultural or elementary occupations. Exposure is also concentrated among the few who have it as the most exposed ten percent of workers account for 36 percent of total exposure in the pooled sample.

Table 2 reports employment shares and mean exposure for each ISCO major group. The exposure gradient is steep and follows a white-collar ordering. Clerical, technical, and professional occupations have the highest exposure, with clerical workers at 0.605, but together they are a small share of employment. The two largest groups, skilled agriculture (54 percent of workers) and elementary occupations (12 percent), have the lowest exposure, at 0.054 and 0.030. The combination of a steep occupational gradient and an employment distribution that is heavily weighted toward its low-exposure end is what produces a low and concentrated aggregate.

Table 3 reports exposure by employment status. Agricultural workers, who are 58.6 percent of the pooled workforce, have a mean exposure of 0.050. Non-agricultural workers are more exposed, but within the non-agricultural sector, wage employees and own-account workers are exposed to exactly the same degree, both at 0.142. The insulating margin is therefore agriculture versus the rest of the economy.

Among non-agricultural wage workers, for whom formality can be measured, informal workers (no written contract, pension, or tax withholding) have a mean exposure of 0.112

Table 2: Employment share and mean AI exposure by ISCO major group (pooled, 23,444 workers)

ISCO major group	Share of workers	Mean exposure
Clerical support	0.7%	0.605
Technicians and associates	1.8%	0.361
Managers	5.1%	0.210
Professionals	4.6%	0.172
Service and sales	12.6%	0.143
Plant and machine operators	3.6%	0.143
Craft and trades	5.4%	0.051
Skilled agriculture	54.0%	0.054
Elementary occupations	12.1%	0.030

Exposure is the task-weighted share of an occupation’s tasks that current generative models can perform, on a zero-to-one scale. Shares are of all workers with a coded occupation in the pooled four-country sample.

Table 3: Employment share and mean AI exposure by employment status (pooled)

Employment status	Share of workers	Mean exposure
Agriculture	58.6%	0.050
Self-employed, non-agriculture	36.0%	0.142
Wage employee, non-agriculture	3.5%	0.142
Other non-agriculture	1.9%	0.163

Status is anchored on the occupation. Agriculture is read from the ISCO major group, and remaining workers are split by the survey’s main-job employer question. The “other” category is the Ethiopian non-agricultural workers, for whom only the one-digit occupation is available.

and formal workers 0.100. If anything, informal wage workers are marginally more exposed, because informal wage work in these economies includes clerical and sales activity that generative models can perform. Geography and sex add little. Urban workers average 0.095 against 0.087 for rural workers, and men 0.092 against 0.085 for women. The concentration of exposure is occupational, and the occupational structure, not residence or contract type, is what places most workers out of reach of the technology.

4 From within-country to cross-country

Table 4 reports, for each of the four micro countries, the mean worker-level exposure measured and an independent country-level measure of AI substitution exposure drawn from a 32-country cross-section of African economies. The worker-level levels are not perfectly

comparable across countries, because the Ethiopian occupation code is coarse and its sample selected, which inflates its measured level. The country-level measure, which does not share that selection, places Ethiopia among the least exposed, consistent with its very high agricultural share.

Table 4: Worker-level and country-level AI exposure, four micro-data countries

Country	Worker-level mean	Country-level exposure
Nigeria 2023	0.122	0.333
Ethiopia 2021	0.144	0.221
Tanzania 2020	0.083	0.238
Uganda 2019	0.055	0.214

The worker-level mean is the pooled automation-exposure score from the LSMS sample. The country-level measure is an independent index of occupational substitution exposure aggregated to the national level.

Across the full 32-country cross-section, the same insulation logic that operates within countries operates between them. Country-level AI exposure falls with the informal employment rate (Spearman rank correlation of -0.74 , 28 countries) and with the agricultural self-employment share (-0.75 , 30 countries), and rises with income per capita ($+0.83$, 31 countries). Figure 1 plots exposure against the informal employment rate and marks the four countries observed at the worker level as they lie along the regional gradient. The cross-country relationship is descriptive, but it confirms that the structural feature shown with microdata, the confinement of exposure to a small non-agricultural minority, is what drives the well-known result that poorer and more informal economies appear less exposed.

5 Limitations

First, the exposure measure is built on US task content and applied to African occupations through a crosswalk. It captures the task composition of an occupation, not local differences in how that occupation is performed, so it is best read as a measure of an occupation’s intrinsic exposure rather than of realised local risk. Second, cross-country comparisons of the cardinal level are imperfect. Ethiopian occupation is coded only at the major-group level, and the Tanzanian module originally omitted subsistence farmers, whom I added at the agricultural level. The conclusions rest on the occupational gradient and on the pooled distribution, rather than on the precise national means. Third, the analysis is descriptive. The study measures exposure, not impact, and a companion comparison of a Nigerian worker panel between 2018 and 2023 shows no exposure-related change in employment over a window

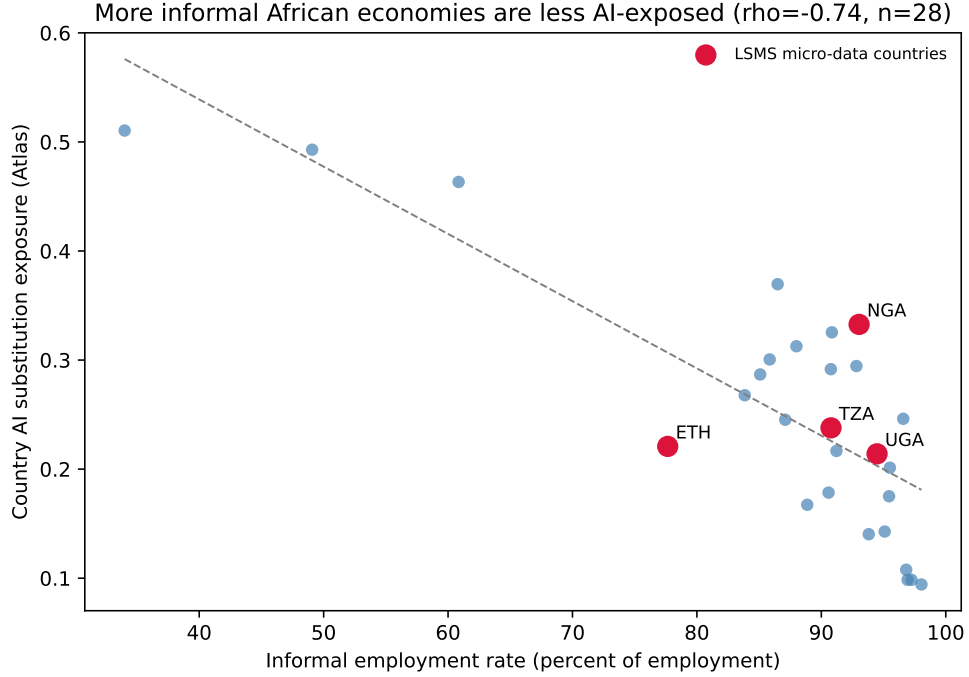


Figure 1: Country-level AI exposure against the informal employment rate, 32 African economies. The four LSMS micro-data countries are marked.

in which generative AI had barely been adopted, so this cross-section is best read as a pre-impact baseline rather than as evidence of an effect.

6 Conclusion

Using LSMS-ISA microdata that include the agricultural and informal workforce, I measure generative-AI exposure across the occupational structure of four Sub-Saharan African economies. Exposure is low, steeply graded across occupations, and concentrated in a small white-collar minority. The agricultural and own-account majority is insulated by the nature of its work rather than by the informality of its contracts. The same structure scales to a 32-country regional gradient in which more informal and more agricultural economies are less exposed. For policy and for measurement, the implication is that aggregate exposure statistics understate how unequally the technology reaches workers in these economies. The gains and risks of generative AI, for now, are confined to the thin formal-sector top of the labour market, while the structural transformation that would broaden exposure is lagging.

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